

Mohit Rana

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INTERNSHIPS

Frontend Developer Intern

June 2024 – Aug 2024

JNS Systech Pvt Ltd

Okhla, New Delhi

- Architected and developed a product listing platform from scratch using **JavaScript, HTML5, and CSS**, supporting **500+ product entries**.
- Implemented dynamic product rendering and optimized DOM manipulation, reducing page load time by **30%**.
- Designed high-fidelity animations and interactive transitions, increasing average session duration by **22%**.
- Built fully responsive UI components, improving cross-device compatibility and reducing layout bugs by **40%**.

PROJECTS

Object Detection System (Raspberry Pi 4 | YOLOv8 | VL53L1X ToF LiDAR | Python | OpenCV)

- Engineered a real-time edge AI object detection system on Raspberry Pi 4 (8GB), achieving **20 FPS at 480x320 resolution** using YOLOv8 and OpenCV.
- Improved inference speed by **35%** via frame skipping, model quantization, and modular pipeline design while maintaining **80-92% detection accuracy**.
- Integrated VL53L1X ToF LiDAR sensor with **<1% distance error**, enabling precise depth-aware object localization on resource-constrained hardware.

Movie Recommendation System (Python | Scikit-learn | Pandas | NumPy | Matplotlib)

- Built a recommendation engine on a dataset of **5,000+ movies**, implementing TF-IDF and Count Vectorization for feature extraction.
- Implemented and benchmarked **Content-Based vs Collaborative Filtering**, improving recommendation relevance by **18% (Precision)**.
- Reduced cold-start recommendation latency by **30%** through optimized similarity computation using cosine similarity matrices.

Book Recommender (Python | Scikit-learn | Pandas | NumPy | K-Means Clustering)

- Developed a clustering-based recommendation system on **8,000+ books**, achieving **85% genre alignment** using K-Means clustering.
- Engineered feature pipelines using author, genre, and metadata vectors to generate **610 highly similar recommendations per query**.
- Reduced average user search time by **40%** through optimized clustering and similarity indexing.

SKILLS

Programming Languages: Python, C++, JavaScript, SQL (PostgreSQL), HTML/CSS

Frameworks & Libraries: React, Scikit-learn, Pandas, NumPy, Matplotlib, OpenCV

Tools & Platforms: Git, Linux (Ubuntu), AWS, VS Code, Jupyter Notebook, MATLAB, SciLab, MS Excel

Embedded & Hardware: Arduino, Raspberry Pi

CERTIFICATIONS

- **Linux Certification** Spoken Tutorial, IIT Bombay 2024
- **Python Programming Certification** TechStack Academy 2024
- **RDBMS (PostgreSQL) Certification** Spoken Tutorial, IIT Bombay 2025
- **Arduino Certification** Spoken Tutorial, IIT Bombay 2025

EDUCATION

Bhagwan Parshuram Institute of Technology <i>B.Tech in Electronics and Communication Engineering (CGPA: 7.6/10)</i>	2022 – 2026 New Delhi
Vidya Bharati School <i>Class XII (CBSE) 82%</i>	2020 – 2021 New Delhi
Vidya Bharati School <i>Class X (CBSE) 92%</i>	2018 – 2019 New Delhi